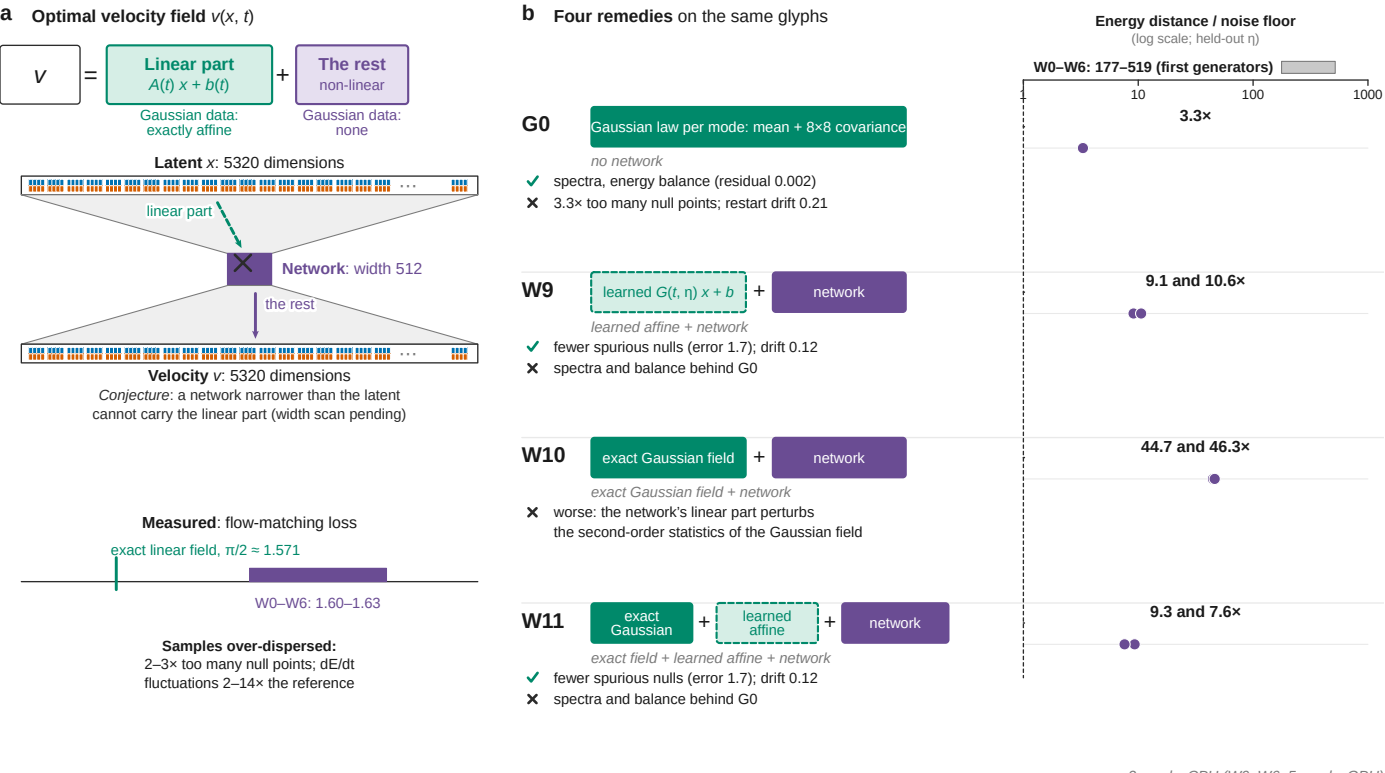

a
Optimal velocity field v(x, t)
v
=
Linear part
A(t) x + b(t)
Gaussian data:
exactly affine
+
The rest
non-linear
Gaussian data:
none
Latent x: 5320 dimensions
linear part
Network: width 512
the rest
Velocity v: 5320 dimensions
Conjecture: a network narrower than the latent
cannot carry the linear part (width scan pending)
Measured: flow-matching loss
exact linear field, π/2 ≈ 1.571
W0–W6: 1.60–1.63
Samples over-dispersed:
2–3× too many null points; dE/dt
fluctuations 2–14× the reference
b
Four remedies on the same glyphs
Energy distance / noise floor
(log scale; held-out η)
W0–W6: 177–519 (first generators)
1
10
100
1000
G0
Gaussian law per mode: mean + 8×8 covariance
3.3×
no network
✓ spectra, energy balance (residual 0.002)
✗ 3.3× too many null points; restart drift 0.21
W9
learned G(t, η) x + b
+
network
9.1 and 10.6×
learned affine + network
✓ fewer spurious nulls (error 1.7); drift 0.12
✗ spectra and balance behind G0
W10
exact Gaussian field
+
network
44.7 and 46.3×
exact Gaussian field + network
✗ worse: the network's linear part perturbs
the second-order statistics of the Gaussian field
W11
exact Gaussian
+
learned affine
+
network
9.3 and 7.6×
exact field + learned affine + network
✓ fewer spurious nulls (error 1.7); drift 0.12
✗ spectra and balance behind G0
2 seeds, CPU (W0–W6: 5 seeds, GPU)